

## Lab Task: Threads

---

### Lab Task 1: Multi-Threaded Student Grade Processing

♦ **Scenario:** A university system needs to **automate grade calculation** for multiple students using threads.

#### Thread Functions:

1. **Thread 1** → Takes student marks as input and assigns a grade based on conditions.
2. **Thread 2** → Finds the highest and lowest marks among students.
3. **Thread 3** → Calculates the overall class percentage.

#### Expected Output Example:

```
Thread 1: Enter marks for 3 students: 85 90 76
Thread 1: Grades assigned -> Student 1: A, Student 2: A+, Student 3: B
Thread 2: Highest marks = 90, Lowest marks = 76
Thread 3: Class percentage = 83.67%
Grade processing completed using multithreading!
```

---

### Lab Task 2: Multi-Threaded Number Processing System

♦ **Scenario:** A system is designed to **process numbers** using multiple threads to improve efficiency.

### Thread Functions:

- 1.Thread 1 → Generates the first 5 even numbers and prints them.
- 2.Thread 2 → Finds the sum of numbers from 1 to 10.
- 3.Thread 3 → Computes the product of the first 5 positive integers.

### Expected Output:

```
Thread 1: First 5 even numbers: 2 4 6 8 10
Thread 2: Sum of numbers from 1 to 10 = 55
Thread 3: Product of first 5 positive integers = 120
Number processing completed using multithreading!
```

---

## Lab Task 3: Multi-Threaded Ticket Booking System (Using `pthread_create()`, `pthread_self()`, and `pthread_exit()`)

♦ **Scenario:** A movie ticket booking system is automated with **multiple threads**, each handling different customers independently.

### Thread Functions:

- 1.Thread 1 → Books a ticket for Customer 1 and prints its thread ID.
- 2.Thread 2 → Books a ticket for Customer 2 and exits.
- 3.Thread 3 → Books a ticket for Customer 3 and confirms their seat.

### Expected Output:

```
Thread 1: Booking ticket for Customer 1, Thread ID:
140439869986560
Thread 2: Booking ticket for Customer 2, exiting...
Thread 3: Booking ticket for Customer 3, Seat
Confirmed!
All ticket bookings completed!
```

---

## Instructions for Submission

- Ensure your code is **well-documented with comments**.
- Implement **proper error handling** Test your program with **various inputs** before submission.
- Submit your **source code along with screenshots of successful runs**.

---

THE END

---